

Technical Support Ticket Management System

MÓDULO 6 — Flujo Real del Sistema de Soporte

Objetivo: Implementar la interactividad para el cambio de estados y la asignación de responsables, cerrando el ciclo de vida del ticket.

En este **MÓDULO 6** se transforma la base de datos en una aplicación con **lógica de aplicación real**. Se implementa el flujo de trabajo: un empleado genera un ticket, el Administrador asigna el ticket a un técnico y el Técnico lo resuelve.

1. Consolidación Final del `TicketService.cs`

Para evitar errores de "método no encontrado", asegurarse de que los servicios en `/Services/TicketService.cs` contengan todos estos métodos clave para la gestión. Se usa la potencia de **LINQ** y **Entity Framework Core 10**. Editar:

```
1  using Microsoft.EntityFrameworkCore;
2  using SupportTicketsSystem.Data;
3  using SupportTicketsSystem.Enums;
4  using SupportTicketsSystem.Models;
5  using System.Security.Claims;
6
7  namespace SupportTicketsSystem.Services;
8
9  // Usando Primary Constructor de C# 14
10 public class TicketService(IDbContextFactory<ApplicationDbContext> dbFactory)
11 {
12     // --- MÓDULO 4: MÉTODOS DE CREACIÓN ---
13
14     1 reference
15     public async Task<List<Category>> GetCategoriesAsync()
16     {
17         using var context = dbFactory.CreateDbContext();
18         return await context.Categories.ToListAsync();
19     }
20
21     1 reference
22     public async Task CreateTicketAsync(Ticket ticket)
23     {
24         using var context = dbFactory.CreateDbContext();
25         context.Tickets.Add(ticket);
26         await context.SaveChangesAsync();
27     }
28
29     // --- MÓDULO 5: SEGURIDAD Y FILTRADO ---
30
31     0 references
32     public async Task<List<Ticket>> GetTicketsForUserAsync(ClaimsPrincipal user)
33     {
34         using var context = dbFactory.CreateDbContext();
35         var userId = user.FindFirst(ClaimTypes.NameIdentifier)?.Value;
36
37         var query = context.Tickets
38             .Include(t => t.Category)
39             .Include(t => t.Author)
40             .Include(t => t.Technician)
41             .AsQueryable();
42
43         if (user.IsInRole(Roles.Admin))
44         {
45             return await query.ToListAsync();
46         }
47         else if (user.IsInRole(Roles.Technician))
48         {
49             return await query.Where(t => t.TechnicianId == userId || t.Status == Enums.TicketStatus.Open).ToListAsync();
50         }
51     }
52 }
```

```

48         return await query.Where(t => t.AuthorId == userId).ToListAsync();
49     }
50
51
52     // --- MÓDULO 6: GESTIÓN ADMINISTRATIVA ---
53
54     0 references
55     public async Task<List<Ticket>> GetAllTicketsAsync()
56     {
57         using var context = dbFactory.CreateDbContext();
58         return await context.Tickets
59             .Include(t => t.Category)
60             .Include(t => t.Author)
61             .Include(t => t.Technician)
62             .OrderByDescending(t => t.CreatedAt)
63             .ToListAsync();
64
65     0 references
66     public async Task UpdateTicketAsync(Ticket ticket)
67     {
68         using var context = dbFactory.CreateDbContext();
69         ticket.UpdatedAt = DateTime.UtcNow;
70         context.Entry(ticket).State = EntityState.Modified;
71         await context.SaveChangesAsync();
72     }
73
74     0 references
75     public async Task<List<ApplicationUser>> GetUsersByRoleAsync(string roleName)
76     {
77         using var context = dbFactory.CreateDbContext();
78
79         // Buscamos el ID del rol primero
80         var role = await context.Roles.FirstOrDefaultAsync(r => r.Name == roleName);
81         if (role == null) return new List<ApplicationUser>();
82
83         // Buscamos los IDs de usuarios asociados a ese rol
84         var userIds = await context.UserRoles
85             .Where(ur => ur.RoleId == role.Id)
86             .Select(ur => ur.UserId)
87             .ToListAsync();
88
89         // Retornamos la lista de objetos de usuario
90         return await context.Users
91             .Where(u => userIds.Contains(u.Id))
92             .ToListAsync();
93
94     // Obtener un ticket específico con todos sus detalles y relaciones
95     0 references
96     public async Task<Ticket?> GetTicketByIdAsync(int id)
97     {
98         using var context = dbFactory.CreateDbContext();
99         return await context.Tickets
100             .Include(t => t.Category)
101             .Include(t => t.Author)
102             .Include(t => t.Technician)
103             .FirstOrDefaultAsync(t => t.Id == id);
104     }
105
106     // Listar todos los técnicos disponibles para asignar
107     0 references
108     public async Task<List<ApplicationUser>> GetTechniciansAsync()
109     {
110         using var context = dbFactory.CreateDbContext();
111         var techRole = await context.Roles.FirstOrDefaultAsync(r => r.Name == Roles.Technician);
112         if (techRole == null) return [];
113
114         var techIds = await context.UserRoles
115             .Where(ur => ur.RoleId == techRole.Id)
116             .Select(ur => ur.UserId).ToListAsync();
117
118         return await context.Users.Where(u => techIds.Contains(u.Id)).ToListAsync();
119
120     // Actualizar el ticket (Asignación o Cambio de Estado)
121     0 references
122     public async Task UpdateTicketStatusAsync(int ticketId, TicketStatus newStatus, string? technicianId = null)
123     {
124         using var context = dbFactory.CreateDbContext();
125         var ticket = await context.Tickets.FindAsync(ticketId);
126         if (ticket != null)
127         {
128             ticket.Status = newStatus;
129             if (!string.IsNullOrEmpty(technicianId)) ticket.TechnicianId = technicianId;
130             ticket.UpdatedAt = DateTime.UtcNow;
131             await context.SaveChangesAsync();
132         }
133     }

```

2. Dashboard Administrativo Básico

Crear el folder /Components/Pages/**Admin**

Crear el archivo /Components/Pages/Admin/**Dashboard.razor**. Aquí el administrador verá el estado del sistema.

```
1  @page "/admin/dashboard"
2  @using Microsoft.AspNetCore.Authorization
3  @using SupportTicketsSystem2026.Data
4  @using SupportTicketsSystem2026.Models
5  @using SupportTicketsSystem2026.Services
6  @using SupportTicketsSystem2026.Enums
7  @inject TicketService TicketService
8  @inject NavigationManager Navigation
9  @attribute [Authorize(Roles = Roles.Admin)]
10 @rendermode InteractiveServer
11
12 <PageTitle>Admin Dashboard</PageTitle>
13
14 <div class="container-fluid mt-4">
15     <div class="d-flex justify-content-between align-items-center mb-4">
16         <h3><i class="bi bi-speedometer2"></i> Administrative Dashboard</h3>
17     </div>
18
19     @* --- TABLA DE GESTIÓN (MÓDULO 6) --- *@
20     <div class="card shadow-sm border-0">
21         <div class="card-header bg-white py-3">
22             <h5 class="mb-0 text-secondary">All Support Requests</h5>
23         </div>
24         <div class="table-responsive">
25             <table class="table table-hover align-middle mb-0">
26                 <thead class="table-light">
27                     <tr>
28                         <th>Title</th>
29                         <th>Author</th>
30                         <th>Category</th>
31                         <th>Status</th>
32                         <th>Technician</th>
33                         <th class="text-center">Actions</th>
34                     </tr>
35                 </thead>
36                 <tbody>
37                     @foreach (var ticket in allTickets)
38                     {
39                         <tr>
40                             <td class="fw-bold">@ticket.Title</td>
41                             <td>@ticket.Author?.Email</td>
42                             <td><span class="badge rounded-pill bg-light text-dark border">
43                                 @ticket.Category?.Name</span>
44                             </td>
45                             <td>
46                                 <span class="badge bg-info">@ticket.Status</span>
47                             </td>
48                             <td>
49                                 @if (ticket.Technician != null)
50                                 {
51                                     <span class="text-primary"><i class="bi bi-person-check"></i>
52                                         @ticket.Technician.Email
53                                     </span>
54                                 }
55                                 else
56                                 {
57                                     <span class="text-muted italic">Unassigned</span>
58                                 }
59                             </td>
60                             <td class="text-center">
61                                 <button class="btn btn-sm btn-primary shadow-sm"
62                                     @onclick="@(() => Navigation.NavigateTo($"/tickets/manage/{ticket.Id}"))">
63                                     <i class="bi bi-gear-fill"></i> Manage
64                                 </button>
65                             </td>
66                         </tr>
67                     }
68                 </tbody>
69             </table>
70         </div>
71     </div>
72 </div>
73
74 @code {
75     private List<Ticket> allTickets = new();
76
77     protected override async Task OnInitializedAsync()
78     {
79         // Cargar Lista
80         allTickets = await TicketService.GetAllTicketsAsync();
81     }
82
83     private void SelectTicket(Ticket ticket)
84     {
85         // Aquí redigiremos a una pagina de gestión del ticket, donde el admin podrá editar el ticket,
86         // asignar técnicos, cambiar estado, etc.
87     }
88 }
89
```

3. Página de Gestión: ManageTicket.razor

Crear el archivo /Components/Pages/Tickets/ManageTicket.razor. Esta página permitirá al Admin asignar al Técnico para trabajar el ticket.

```
1  @page "/tickets/manage/{TicketId:int}"
2  @using Microsoft.AspNetCore.Authorization
3  @using SupportTicketsSystem2026.Data
4  @using SupportTicketsSystem2026.Models
5  @using SupportTicketsSystem2026.Enums
6  @using SupportTicketsSystem2026.Services
7  @inject TicketService TicketService
8  @inject NavigationManager Navigation
9  @attribute [Authorize(Roles = $"{Roles.Admin},{Roles.Technician}")]
10 @rendermode InteractiveServer
11
12 <div class="container mt-4">
13     @if (ticket == null)
14     {
15         <p><em>Loading...</em></p>
16     }
17     else
18     {
19         <div class="card shadow">
20             <div class="card-header bg-primary text-white d-flex justify-content-between">
21                 <h4 class="mb-0">Manage Ticket #@ticket.Id</h4>
22                 <span class="badge bg-light text-dark">@ticket.Status</span>
23             </div>
24             <div class="card-body">
25                 <h5>@ticket.Title</h5>
26                 <p class="text-muted">
27                     Reported by: @ticket.Author?.Email on @ticket.CreatedAt.ToShortDateString()
28                 </p>
29                 <hr />
30                 <p>@ticket.Description</p>
31
32                 <div class="row mt-4 bg-light p-3 rounded">
33                     @* SECCIÓN PARA EL ADMINISTRADOR: ASIGNACIÓN *@
34                     <AuthorizeView Roles="@Roles.Admin">
35                         <div class="col-md-6 border-end">
36                             <h6>Assign Technician</h6>
37                             <select class="form-select mb-2" @bind="selectedTechId">
38                                 <option value="">-- Select Technician --</option>
39                                 @foreach (var tech in technicians)
40                                 {
41                                     <option value="@tech.Id">@tech.Email</option>
42                                 }
43                             </select>
44                             <button class="btn btn-success w-100" @onclick="AssignTicket">
45                                 Confirm Assignment
46                             </button>
47                         </div>
48                     </AuthorizeView>
49
50                     @* SECCIÓN PARA EL TÉCNICO: CAMBIO DE ESTADO *@
51                     <div class="col-md-6">
52                         <h6>Update Status</h6>
53                         <div class="d-grid gap-2 d-md-block">
54                             <button class="btn btn-outline-warning" @onclick="() =>
55                                 UpdateStatus(TicketStatus.InProgress)">Start Work</button>
56                             <button class="btn btn-outline-success" @onclick="() =>
57                                 UpdateStatus(TicketStatus.Resolved)">Mark Resolved</button>
58                         </div>
59                     </div>
60                 </div>
61             </div>
62         </div>
63     }
64 </div>
65 @code {
66     [Parameter] public int TicketId { get; set; }
67     private Ticket? ticket;
68     private List<ApplicationUser> technicians = [];
69     private string selectedTechId = "";
70
71     protected override async Task OnInitializedAsync()
72     {
73         ticket = await TicketService.GetTicketByIdAsync(TicketId);
74         technicians = await TicketService.GetTechniciansAsync();
75         if (ticket?.TechnicianId != null) selectedTechId = ticket.TechnicianId;
76     }
77
78     private async Task AssignTicket()
79     {
80         if (string.IsNullOrEmpty(selectedTechId)) return;
81         await TicketService.UpdateTicketStatusAsync(TicketId, TicketStatus.InProgress, selectedTechId);
82         Navigation.NavigateTo("/admin/dashboard");
83     }
84
85     private async Task UpdateStatus(TicketStatus status)
86     {
87         await TicketService.UpdateTicketStatusAsync(TicketId, status);
88         ticket = await TicketService.GetTicketByIdAsync(TicketId); // Recargar datos
89     }
90 }
```

Ejecute la Aplicación. Debe cargar sin errores, debe poder hacer login, registrar usuarios.

Actualizando el Repositorio en GitHub (requerido)

```
Developer PowerShell

+ Developer PowerShell v |  | 
*****
** Visual Studio 2026 Developer PowerShell v18.4.1
** Copyright (c) 2026 Microsoft Corporation
*****
PS C:\Users\luisd\source\repos\SupportTicketsSystem2026> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   SupportTicketsSystem2026/Services/TicketService.cs

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        SupportTicketsSystem2026/Components/Pages/Admin/
        SupportTicketsSystem2026/Components/Pages/Tickets/ManageTicket.razor

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\luisd\source\repos\SupportTicketsSystem2026>
PS C:\Users\luisd\source\repos\SupportTicketsSystem2026> git add .
PS C:\Users\luisd\source\repos\SupportTicketsSystem2026> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   SupportTicketsSystem2026/Components/Pages/Admin/Dashboard.razor
        new file:   SupportTicketsSystem2026/Components/Pages/Tickets/ManageTicket.razor
        modified:   SupportTicketsSystem2026/Services/TicketService.cs

PS C:\Users\luisd\source\repos\SupportTicketsSystem2026> git commit -m "Mdulo 6: Implementando la logica del Flujo Real del Sistema de Soporte."
[main 2bc148a] Mdulo 6: Implementando la logica del Flujo Real del Sistema de Soporte.
 3 files changed, 263 insertions(+)
 create mode 100644 SupportTicketsSystem2026/Components/Pages/Admin/Dashboard.razor
 create mode 100644 SupportTicketsSystem2026/Components/Pages/Tickets/ManageTicket.razor
PS C:\Users\luisd\source\repos\SupportTicketsSystem2026> git push origin main
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 24 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (11/11), 4.49 KiB | 2.24 MiB/s, done.
Total 11 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/siciuprp2020/SupportTicketsSystem2026.git
 5510d51..2bc148a  main -> main
PS C:\Users\luisd\source\repos\SupportTicketsSystem2026>
```

